

RESIDUALS, r^2 & MODEL FIT

new 5.4–5.5 • old 2.7–2.8 • ★ old 2.9, 9.1–9.5

ZONE A · THE FORMULAS + THE PLOTS

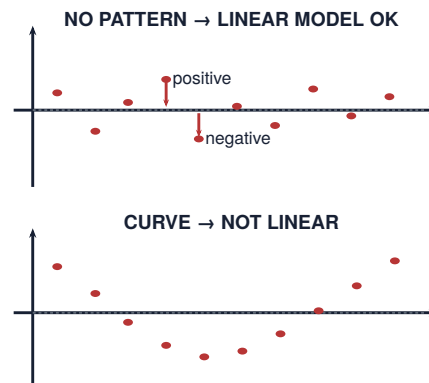
$$\text{residual} = y - \hat{y} = \text{actual} - \text{predicted}$$

r^2 = fraction of the variation in y explained by the linear relationship with x

$$s = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n - 2}}$$

s is the **typical size of a residual**. It has the same units as the response variable.

READ THE RESIDUAL PLOT



ZONE B · COMPUTER-OUTPUT DECODER

Predictor	Coef	SE Coef	T	P	Read it as
Constant	12.41	1.86	6.67	<.001	→ a , the intercept
Study hours	4.83	0.72	6.71	<.001	→ b , s_b , t , P -value
$S = 3.17$		→ s	$R\text{-Sq} = 78.9\%$ → r^2		

ZONE C · SAY IT IN WORDS

"About _____% of the variation in _____ is explained by the linear relationship with _____."

"The actual _____ was _____ units _____ what the line predicted."

ZONE D · WATCH OUT + BEYOND

MODEL-FIT RULES

No extrapolation: do not predict beyond the observed x -range.

r^2 gives the explained fraction, not the direction. Use r for direction.

★ BEYOND THE EXAM

Departures: outlier · high leverage · influential point

$$\log(\hat{y}) = a + bx \Rightarrow \hat{y} = 10^{a+bx}$$

Slope inference ($df = n - 2$):

$$\mu_b = \beta \quad \sigma_b = \frac{s}{\sigma_x \sqrt{n}}$$

$$s_b = \frac{s}{s_x \sqrt{n-1}} \quad t = \frac{b}{s_b} \quad b \pm t^* s_b$$